

Practical # 6

Write a program which uses arithmetic operators to calculate the area of triangle and volume of sphere.

```
#include <stdio.h>

int main() {
    float base, height, radius;
    float area, volume;

    // Area of triangle
    printf("Enter base and height of triangle: ");
    scanf("%f %f", &base, &height);
    area = 0.5 * base * height;
    printf("Area of Triangle = %.2f\n", area);

    // Volume of sphere
    printf("Enter radius of sphere: ");
    scanf("%f", &radius);
    volume = (4.0 / 3.0) * 3.14159 * radius * radius * radius;
    printf("Volume of Sphere = %.2f\n", volume);

    return 0;
}
```

```
Enter base and height of triangle: 3 4
Area of Triangle = 6.00
Enter radius of sphere: 9
Volume of Sphere = 3053.63
```

Algorithm:

1. Start
2. Declare floating-point variables for base, height, radius, area, and volume
3. Prompt the user to enter the base and height of the triangle
4. Read the base and height values
5. Calculate the area of the triangle using the formula
6. $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$
7. Display the calculated area of the triangle
8. Prompt the user to enter the radius of the sphere
9. Read the radius value
10. Calculate the volume of the sphere using the formula
11. $\text{Volume} = \left(\frac{4}{3}\right) \times \pi \times \text{radius}^3$
12. Display the calculated volume of the sphere
13. End